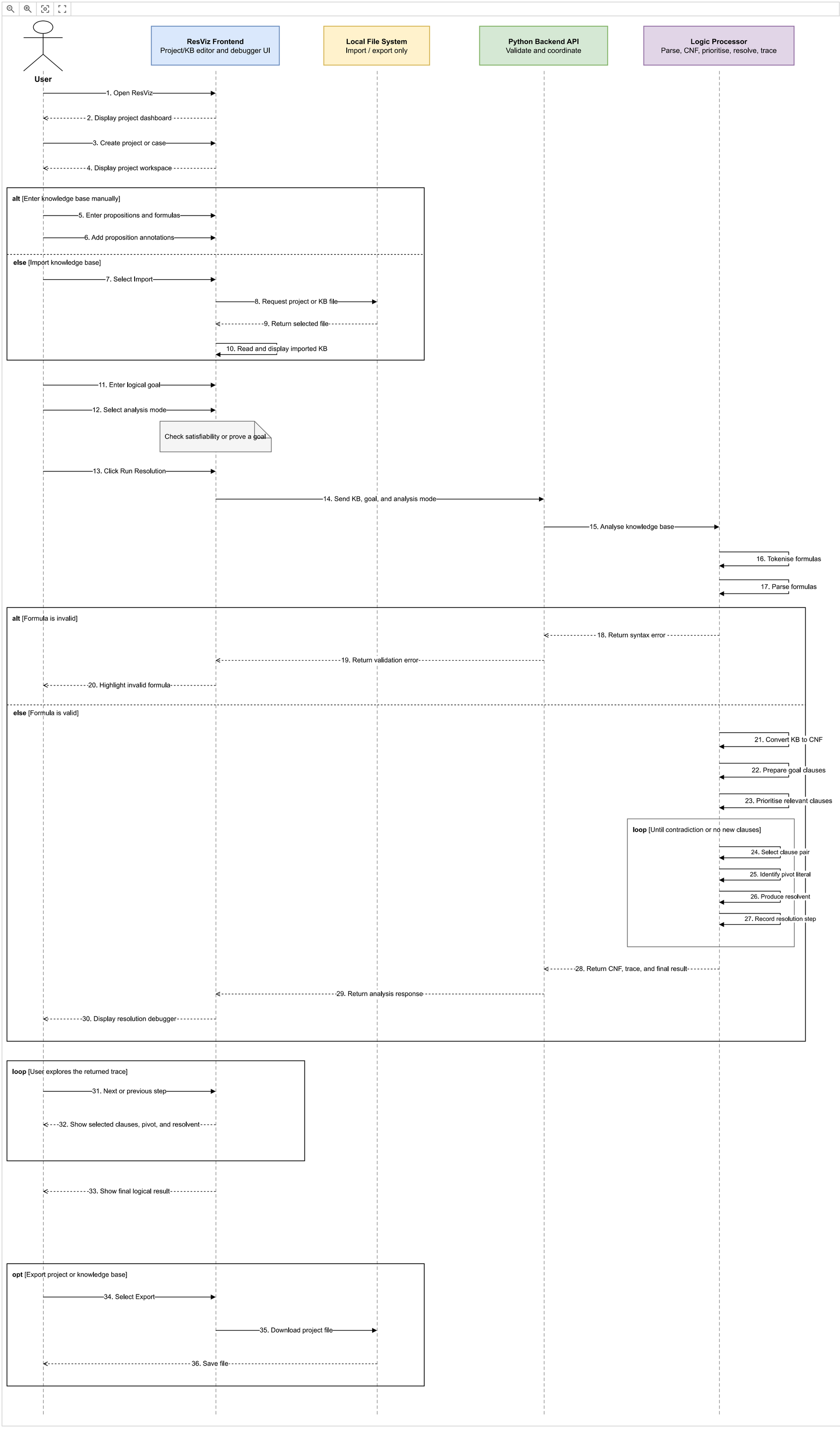

User
ResViz Frontend
Project/KB editor and debugger UI
Local File System
Import / export only
Python Backend API
Validate and coordinate
Logic Processor
Parse, CNF, prioritise, resolve, trace
1. Open ResViz
2. Display project dashboard
3. Create project or case
4. Display project workspace
alt [Enter knowledge base manually]
5. Enter propositions and formulas
6. Add proposition annotations
else [Import knowledge base]
7. Select Import
8. Request project or KB file
9. Return selected file
10. Read and display imported KB
11. Enter logical goal
12. Select analysis mode
Check satisfiability or prove a goal
13. Click Run Resolution
14. Send KB, goal, and analysis mode
15. Analyse knowledge base
16. Tokenise formulas
17. Parse formulas
alt [Formula is invalid]
18. Return syntax error
19. Return validation error
20. Highlight invalid formula
else [Formula is valid]
21. Convert KB to CNF
22. Prepare goal clauses
23. Prioritise relevant clauses
loop [Until contradiction or no new clauses]
24. Select clause pair
25. Identify pivot literal
26. Produce resolvent
27. Record resolution step
28. Return CNF, trace, and final result
29. Return analysis response
30. Display resolution debugger
loop [User explores the returned trace]
31. Next or previous step
32. Show selected clauses, pivot, and resolvent
33. Show final logical result
opt [Export project or knowledge base]
34. Select Export
35. Download project file
36. Save file